

Separating concerns in Jim Sanborn's Kryptos

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Abstract

This paper details four discrete layers of information within Jim Sanborn's 1990 Kryptos sculpture. These layers are shown to include a mathematical blueprint for calculating astronomical calendar cycles, a series of static challenge puzzles, two kinetic puzzles and finally a novel classical cipher type known more widely by its key data as K4. The challenge puzzles when solved in order will be shown to act as a guide to discovery and isolation of the K4 cipher. Analysis of the K4 cipher type once isolated is presented and classified as being 'Cymatic' in nature.

1 Supporting information and credits

Much of the contents of this paper have already been published piecemeal over the last 14 years by the author to online forums and social media, but not formally entered into the academic record as a holistic body of work. All results within this paper are thought to be the original work of the author, but credit is given to James Churchill who provided useful information on the topic of 'kinetic art' and further directed attention to a significant set of repeating letter groupings in the Kryptos cipher text in late 2020. It would be difficult to sensibly condense these findings into a concise treatment, so a companion tool written in HTML and JavaScript is available alongside this paper in order to allow reproduction of the results presented. It will be assumed that readers are already familiar with Kryptos and the plain text (pt) results already decrypted or known to date for sections K1, K2, K3 and K4.

1.1 Introduction

Despite some 25% of the K4 pt having been disclosed by Jim Sanborn since 2010 there remains confusion as to what Kryptos is and

whether or not cryptanalysis can be applied to K4 as was the case for sections K1-3. This paper looks to separate the various concerns that contend for attention in the overall sculpture, so that K4 can be fully isolated with a high degree of confidence. It is hoped that isolation of the cipher will finally allow for a proper cryptanalysis. To achieve the level of isolation required for such an analysis, four frameworks that exist in superposition are presented for logical separation. Section 2 deals with an 'implied scheme' hidden via steganography that exposes a mathematical blueprint for a number of well known calculations relating to the Solar Lunar cycle. These cycles are in turn shown to be analogous to the functionality of the Antikythera Mechanism. Section 3 details a framework predicated on four six letter keys in the pt that lay out a trail of challenge puzzles each of varying difficulty and significance, that once solved in order lead to the identification of 6 matrices upon which the K4 cipher is predicated before K4 per se can be used as a key for pt retrieval. Section 4 will show the extraction of additional information via a kinetic scheme that requires physical interaction with discrete rows of Kryptos cipher text (ct). This kinetic aspect further informs understanding of the K4 ciphers mechanical aspects and reveals an analogue already present in the pt. Finally, the nature of the isolated K4 framework will be shown to be a novel form of classical cipher that relies on a phenomenon akin to wave interference and as a result has been given a classification of being 'Cymatic'.

2 Analogue Computation of K1 and K2

Publicly available documents prior to the official dedication of Kryptos on November 3rd 1990 show that Jim Sanborn had already written to employees on the CIA campus before the installation was complete telling them that parts

of Kryptos would be encoded using a Vigenère tableau. Examining the layout of these two tableau panels and laying them out in a fixed width font shows that all letters fit within a 31 x 28 grid with the exception of an additional L that sits alone in column 32. Within this grid an inner 26 x 26 letter square from column 2 row 3 to column 27 row 28 can be formed as a Vigenère table. Such a table is sufficient for the purposes of decrypting K1 and K2 using the keywords PALIMPSEST and ABSCISSA respectively. Performing these two decryptations using a Vigenère table rather than a decryption function running in software means the solve is actually being undertaken using a basic form of analogue computation. This point appears to have been overlooked to date, but it is important as the residual mask left by these two decryption operations contains important structural information that can be used to unlock a discrete layer of Kryptos that has been obscured using steganography.

2.1 Artefacts post decryption of K1 & K2

If every letter in the previously defined 26 x 26 square is shaded to show its utilization in both decryption operations, then it will be seen that only 160 characters of the 676 available have actually been used. Examining where the boundary lies between the top and bottom panels it is seen that of the 160 characters used, 132 were highlighted in the upper panel and 28 in the lower.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	
A	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P
B	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T
C	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O
D	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S
E	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A
F	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B
G	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C
H	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D
I	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E
J	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F
K	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G
L	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H
M	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I
N	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J
O	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L
P	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M
Q	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N
R	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q
S	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U
T	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V
U	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W
V	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X
W	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
X	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K
Y	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R
Z	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	

Figure 1: Letters used in the analogue decryption of parts K1 and K2.

2.2 Effects of 90° right rotation

Taking the mask generated from these 160 characters a right hand rotation of 90 degrees can be applied, which can then be overlaid on either the tableau or ct panels. Performing either operation shows that the width of the mask is in fact only 19 characters across, so when laid across the ct panels a region is defined that has a bound of 19x26 on the left and an unbounded region to the right that is 13x28.

E	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D	*
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	*
G	G	W	H	K	K	?	D	Q	M	C	P	F	O	Z	D	Q	M	M	I	A	G	P	F	X	H	Q	R	L	G	*	*
T	I	M	V	M	Z	J	A	N	Q	L	V	K	E	N	J	T	B	J	L	B	Q	C	R	T	U	N	J	D	F	H	R
Q	Z	G	Z	L	E	C	G	Y	U	X	U	E	E	N	J	T	B	J	L	B	Q	C	R	T	U	N	J	D	F	H	R
Y	I	Z	E	T	K	Z	E	M	V	D	U	F	K	S	J	H	K	F	W	H	K	U	M	Q	L	S	Z	F	T	I	*
H	H	D	D	D	U	V	H	?	D	W	K	B	F	U	F	P	W	N	T	D	F	I	Y	C	U	Q	Z	E	R	E	*
E	V	L	D	K	F	E	Z	M	O	O	O	J	L	T	T	U	G	S	Y	Q	P	F	E	U	N	L	A	V	I	D	X
F	L	G	T	E	Z	?	F	K	Z	B	S	F	D	Q	V	G	O	G	I	P	U	F	X	H	H	D	R	K	F	*	
F	H	Q	N	T	G	P	U	A	E	C	N	U	V	P	D	J	M	Q	C	L	Q	U	M	U	N	E	D	F	Q	*	
E	L	Z	Z	V	R	R	G	K	F	F	V	O	E	E	X	B	D	M	V	F	N	F	Q	X	E	Z	L	G	R	E	*
D	N	Q	F	M	P	N	?	G	L	F	L	P	M	R	J	Q	Y	A	L	M	G	N	U	V	P	D	X	V	K	P	*
D	Q	U	M	E	B	E	D	M	H	D	A	F	M	J	G	Z	N	U	P	L	G	E	W	J	L	L	A	E	T	G	*

E	N	D	Y	A	H	R	O	H	N	L	S	R	H	E	O	C	P	T	E	O	I	B	I	D	Y	S	H	N	A	I	A
C	H	T	N	R	E	Y	U	L	D	S	L	S	L	S	L	N	O	H	S	N	O	S	M	R	W	X	M	N	E	*	*
T	P	R	N	G	A	T	I	H	N	R	A	R	P	E	S	L	N	N	E	L	E	B	L	P	I	I	A	C	A	E	*
W	M	T	W	N	D	I	T	E	E	N	R	A	H	C	T	E	N	E	U	D	R	E	T	N	H	A	E	O	E	*	*
T	F	O	L	S	E	D	I	W	E	N	H	A	E	I	O	Y	T	E	Y	Q	H	E	E	N	C	T	A	Y	C	R	*
E	I	F	T	B	R	S	P	A	M	H	H	E	W	E	N	A	T	A	M	A	T	E	G	Y	E	R	L	B	*	*	
T	E	E	F	O	A	S	F	I	O	T	U	E	T	U	A	E	O	T	O	A	R	M	A	E	E	R	T	N	R	T	I
B	S	E	D	D	N	I	A	A	H	T	M	S	T	E	W	P	I	E	R	O	A	G	R	I	E	W	F	E	B	*	
A	E	C	T	D	D	H	I	L	C	E	I	H	S	I	T	E	G	O	E	A	O	S	D	D	R	Y	D	L	O	R	I
R	K	L	M	L	E	H	A	G	T	D	H	A	R	D	P	N	E	O	H	M	G	F	M	F	E	U	H	E	*	*	
E	C	D	M	R	I	P	F	E	I	M	E	H	N	L	S	S	T	T	R	T	V	D	O	H	W	?	O	B	K	R	*
U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O	*
T	W	T	M	Z	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P	*
V	T	T	Q	S	J	Q	S	E	K	D	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	U	E	K	C	A	R	*

Figure 2: 90° right hand rotation of decryption mask overlaying the ct panels.

Post rotation and overlay, the distribution of highlighted characters shown above are now 97 in the upper panel and 63 in the lower, which are the respective lengths of parts K4 and K1.

2.3 Layout of the top left panel

The overlay of the decryption mask detailed in Figure 2 exposes an underlying numerical canvas hidden via steganography that is independent of the ct itself and relies solely on the position of the 3 question mark characters and the presence of one additional character that sits alone in the 33rd column of row 23. Attention is initially drawn to this scheme by the repeating rows of 19 in the rotated decryption mask. These 3 initial rows of 19 blocks can be summed to give values of 38 and 57. These numbers are the degrees and minutes of the northerly coordinates given in the pt of K2. Looking at the location of the ? characters in relation to these repeating rows of 19 a complete numerical description can be derived for all of the given coordinates for the sculpture, which are [38, 57, 6.5] and [77, 8, 44]. Following the coordinate blocks the numbers 3, 12 and 7 are defined, which

can be used to approximate the transcendental numbers pi and e. The last two numbers in the scheme are given as 4 and 48. The unbounded region in green on the right in figure 3 is a 13x28 grid with an extra character in column 33 row 23 giving a logical block size of 365 characters.

Y	M	U	F	P	H	Z	L	R	F	A	X	Y	U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J
Y	Q	T	Q	U	X	Q	B	Q	V	Y	U	V	L	L	T	R	E	V	J	Y	Q	T	M	K	Y	R	D	M	F	D	*
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	O	E	T	G	F	Q	J	N	C	E	*
G	G	W	H	K	K	?	D	Q	M	C	P	F	Q	Z	D	Q	M	H	I	A	G	P	F	X	H	Q	R	L	G	*	*
T	I	M	V	M	Z	J	A	N	Q	L	V	K	Q	E	D	A	G	D	V	F	R	P	J	U	N	G	E	U	N	A	*
Q	Z	G	Z	L	E	C	G	Y	U	X	U	E	E	N	J	T	B	J	L	B	Q	C	R	T	B	J	D	F	H	R	*
Y	I	Z	E	T	K	Z	E	M	V	D	U	F	K	S	J	H	K	F	W	H	K	U	W	Q	L	S	Z	F	T	I	*
H	H	D	D	D	U	V	H	?	D	W	K	B	F	U	F	P	W	N	T	D	F	I	Y	C	U	Q	Z	E	R	E	*
E	V	L	D	K	F	E	Z	M	O	Q	Q	J	L	T	T	U	G	S	Y	Q	P	F	E	U	N	L	A	V	I	D	X
F	L	G	G	T	E	Z	?	F	K	Z	B	S	F	D	Q	V	G	O	G	I	P	U	F	X	H	H	D	R	K	F	*
F	H	Q	N	T	G	P	U	A	E	C	N	U	V	P	D	J	M	Q	C	L	Q	U	M	U	N	E	D	F	Q	*	*
E	L	Z	Z	V	R	R	G	K	F	F	V	O	E	E	X	B	D	M	V	P	N	F	Q	X	E	Z	L	G	R	E	*
D	N	Q	F	M	P	N	Z	G	L	F	L	P	M	R	J	Q	Y	A	L	M	G	N	U	V	P	D	X	V	K	P	*
D	Q	U	M	E	B	E	D	M	H	D	A	E	M	J	G	Z	N	U	P	L	G	E	W	J	L	L	A	E	T	G	*

E	N	D	Y	A	H	R	O	H	N	L	S	R	H	E	O	C	P	T	E	O	I	B	I	D	Y	S	H	N	A	I	A	
C	H	T	N	R	E	Y	U	L	D	S	L	S	L	S	L	N	O	H	S	N	O	S	M	R	W	X	M	N	E	*	*	
T	P	R	N	G	A	T	I	H	N	R	A	R	P	E	S	L	N	N	E	L	E	B	L	P	I	I	A	C	A	E	*	
W	M	T	W	N	D	I	T	E	E	N	R	A	H	C	T	E	N	E	U	D	R	E	T	N	H	A	E	O	E	*	*	
T	F	O	L	S	E	D	T	I	W	E	N	H	A	E	I	O	Y	T	E	Y	Q	H	E	E	N	C	T	A	Y	A	C	R
E	I	F	T	B	R	S	P	A	M	H	E	W	E	N	A	T	A	M	A	T	E	G	Y	E	E	R	L	B	*	*	*	
T	E	E	F	O	A	S	F	I	O	T	U	E	T	U	A	E	O	T	O	A	R	M	A	E	E	R	T	N	R	T	I	
B	S	E	D	D	N	I	A	A	H	T	E	M	S	T	E	W	P	I	E	R	O	A	G	R	I	E	W	F	E	B	*	
A	E	C	T	D	D	H	I	L	C	E	I	H	S	I	T	E	G	O	E	A	O	S	D	D	R	Y	D	L	O	R	I	T
R	K	L	M	L	E	H	A	G	T	D	H	A	R	D	E	N	E	O	H	M	G	F	M	F	E	U	H	E	*	*	*	
E	C	D	M	R	I	P	F	E	S	I	M	E	H	N	L	S	S	T	T	R	T	V	D	O	H	W	?	O	B	K	R	*
U	O	X	O	G	H	U	L	B	S	E	P	I	E	B	B	W	F	L	R	V	Q	P	R	N	G	K	S	S	O	*	*	
T	W	T	Q	S	J	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P	*	
V	T	T	M	Z	F	E	K	W	G	D	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	U	E	K	C	A	R	*	

Figure 3: Steganographic scheme showing the coordinates, 5 additional numbers and a 365 block calendar.

2.4 A numerical blueprint for gearing

The numbers given for the degrees and minutes of the K2 coordinates show an intriguing mathematical property, whereby 38, 57, 77 and 8 all sum to 180. Remembering that K1 and K2 were derived by driving the tableau layout with a basic analogue algorithm, it can be seen that the numerical relationships detailed in figure 3 show that 180 blocks of the numerical scheme detailed above could be utilized to drive an analogue cycle in which the 180 blocks represent angular degrees. Traversing this path twice would of course be representative of passing through all the degrees of a circle. Given the presence of a calendar block on the right hand side of the scheme it is reasonable to suggest that the act of passing through 360° may well relate to the passing of a sidereal day. If this is a valid interpretation then an analogue operation is capable of setting off a cursor that continually cycles through the 180 blocks that originally defined the degrees and minutes of the coordinates. Every time 360 blocks have been traversed a second cursor can then be moved one block forward in the calendar section, which gives a basic analogue scheme for

computing the day of the year with respect to the Earth's rotation. With regards leap years, it is already known that an additional character can be added to K2 for the missing X in the pt. Such an addition will push out the space (shown as *) in row 14 in figure 3 to become a 366th constituent of the calendar block. Note: In order for the calendar to accommodate February 29th each leap year (in row 14 column 33), the calendar would need to begin on August 31st.

2.5 Analogue derivation of K3

If the 180 block scheme is extended to traverse an additional 12 blocks corresponding to the number 6.5 in the numeric scheme shown on row 4, then the repeating cycle of 180 will now traverse 192 blocks before starting again. This is a noteworthy result as the position of the moon when observed each day appears to moves 13° east, so every time the earth rotates 180° the moon will have appeared to have moved 6.5° respectively. Using the analogue strategy employed previously for tracking the calendar day in 2.4, another cursor can be set running in K3 that moves in concert with the traversal of the 192 blocks specified above. If the value of the character under the cursor in K3 prints every time the cycle repeats then is will simultaneously decrypt K3 due to its 192 character skip period. Simultaneously, the same algorithm will also be computing the relative motion of the moon in the sky with respect to the Earth's rotation.

2.6 Solar lunar cycles.

The derivation of an algorithm capable of showing the lunar cycle in 2.5 draws attention to the numbers participating in the coordinate scheme as a whole. Given that 38 + 57 + 6 + 5 + 77 + 8 + 44 all sum to 235 it can be inferred in light of the previous lunar context that this is a pointer to the Metonic cycle. The Metonic cycle relates integer numbers of synodic months and solar years in the ratio 235:19. This cycle, whilst reasonably accurate can have its precision improved by being smoothed over 4 Metonic cycles, giving rise to the Callippic super cycle. This Callippic cycle is of interest as the first three full rows of 19 blocks in the numeric scheme would each be capable of storing the 19

years of a single Metonic cycle. The coordinate scheme as a whole has the capacity to track 235 blocks relating to individual synodic months, which allows an analogue algorithm to be devised that is capable of tracking 3 historic Metonic cycles of 19 years independently of the current cycle that can be tracked instead in terms of its 235 months. Such a scheme for recording and tracking periods of time hierarchically is also seen in Berlin's "Set Theory Clock".

2.7 The Saros Cycle

On inspection it can further be shown that the tableau panels also lend themselves to a similar steganographic scheme, but with repeating row widths of 18 blocks rather than 19. The calendar portion is exposed by virtue of the extra L in column 32 and allows for a discrete analogue computation of the saros cycle that can be used for reasonably accurate eclipse prediction. This algorithm will not be detailed here, but the point is made that the presence of two separate calendars, one dealing with the Metonic cycle and the other dealing with the Saros is also seen in the Antikythera mechanism, leading the author to conclude that the steganographic scheme is intended to provide a blueprint for the complete recreation of the Antikythera mechanism.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D		
A	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	
B	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	
C	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	
D	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	
E	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	
F	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	
G	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	
H	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	
I	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	
J	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	
K	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	
L	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	
M	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	
N	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L
O	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M
P	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N
Q	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q
R	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U
S	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V
T	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W
U	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
V	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
W	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R
X	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R
Y	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y
Z	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	

Figure 4: Steganographic scheme showing the Saros calendar embedded in the two tableau panels.

3 Challenge Puzzles

The second discrete layer of Kryptos relates to a series of challenge puzzles that once solved in the correct order lead to the isolation of distinct rows in the ct that have had the K4 cipher scheme

overlaid as a palimpsest. These challenge puzzles vary in difficulty, but have well defined outputs that provide additional information as a reward for their solution. For the purposes of this paper, only the challenges relevant to the isolation of K4 will be highlighted with the intention of showing them solved in the correct order. In the steganographic layer detailed previously the two cipher text panels were considered side by side with the two tableau panels. For all relationships discussed from here on the two panels of cipher text will always be assumed to be side by side with the two panels showing the resulting pt in the same positions. The K4 cipher text will remain unaltered in both views. This requirement to consider the ct and pt side by side simultaneously will now be shown to be key to solving the 4 following puzzle challenges.

3.1 Challenge puzzle 1

The first logical break in the challenge puzzle layer is found by isolation of the cipher and pt characters in rows 15-20 between columns 17-23. This gives rise to two 7x6 rectangles that are shown here simultaneously for the ct and pt, both of which occupy the exact same grid position within the 32x28 layout.

C	P	T	E	O	I	B
N	O	H	S	N	O	S
L	N	N	E	L	E	B
E	N	E	U	D	R	E
O	Y	T	E	Y	Q	H
A	T	A	M	A	T	E

S	L	O	W	L	Y	T
H	A	T	E	N	C	U
D	O	O	R	W	A	Y
A	N	D	S	I	M	A
L	E	F	T	H	A	N
H	O	L	E	A	L	I

Figure 5: Challenge Puzzle 1.

It is seen that the cipher text in row 15 gives an incomplete anagram of COEPTIS whereby the S is substituted for a B and similarly in the pt an incomplete anagram of ANNUIT is given whereby the N is substituted for a Y. Isolating the 6 letters within the middle of the 7x6 pt rectangle reveals an anagram of, "ID ROWS". Given the misspellings are B and Y, this would appear to be a direct reference to the last 8 characters of the unmodified K2 pt, which are ID BY ROWS. It would also imply that the phrase was intentional rather than an error. The phrase "ANNUIT COEPTIS" is of course taken from the reverse side of the Great Seal of the United States. The

The results derived here for challenge puzzle 3 were originally found via brute force computation of pt letter frequencies in the regions defined by the previously identified 3x32 matrices. For the purposes of this paper the solve will proceed instead from information present at the center of another rectangle that participates in the same scheme starting to emerge from the previous two challenges. In keeping with the emerging paradigm starting to take shape, another 7x6 rectangle will be drawn to the left of the one previously described in challenge puzzle 2 in rows 9-14, but this time in columns 8-14. It can be seen that this rectangle has at its center another anagram that this time spells out the partial sentence, “YES ITS”. This is significant as it will be remembered that the entry that proceeds “can

you see anything” in Howard Carter’s diary reads, “Yes, it is wonderful”.



Figure 7: Challenge puzzle 3 shown relative to challenge puzzle 2 in the pt view.

Comparing the rows in the pt that contain the letters required to make the string “YES ITS” to those rows participating in the 6 matrices in the ct previously identified, it becomes apparent that there is a strong correlation, with only rows 10 and 27 fulfilling the minimum letter condition, but not falling across a 3x32 matrix position. As a result, these 2 rows have been removed from consideration in figure 8. After analysis, it also becomes apparent that the letter E, which is shown in yellow in figure 8 appears to be participating in a rudimentary scheme that allows the 3x32 matrices to be labelled in a set order.

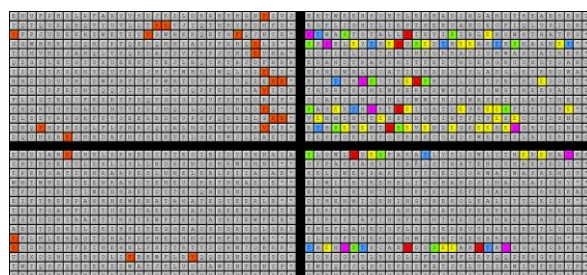


Figure 8: Simultaneous view of ct and pt showing pt rows containing “YES ITS” relative to ct rows marked with F, RE and V.

This scheme is obscured at first glance by two factors. The first is that there are two qualifying rows that fall across the top and bottom rows of the matrix in rows 11 -13. As a result of this the frequency of the letter E from row 12 that sits between them is taken instead. The second obscuration is the word “DESPARATLY” in the pt of row 15, which has had one E altered and one removed. If these are reintroduced, then row 15 now has a frequency count of 5 for the letter E. This allows for labelling of each individual matrix

marked by an F, RE and V with a distinct number from 1-6, the results of which can be tabulated as follows.

Matrix Number	CT Rows
1	1 – 3
2	7 - 9
3	24-26
4	11 – 13
5	13 - 15
6	3 - 5

Table 1: Ct 3x32 matrices marked post labelling.

3.4 Challenge puzzle 4

There is a final challenge puzzle that completes the symmetry of the scheme presented so far and fills in the missing 7x6 rectangle that would logically create a complete and symmetrical set of four. This is shown in figure 9 and is an anagram of “TOMB LR”.

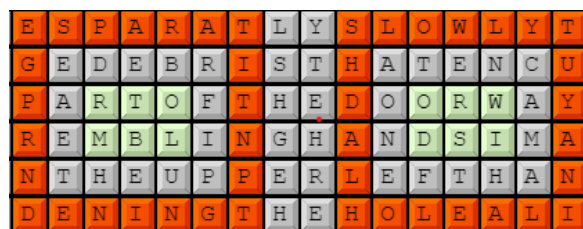


Figure 9: Challenge puzzle 4 shown relative to the position of challenge puzzle 1.

Only two instances of the letters for TOMB exist in the pt that have a contiguous grouping and which have 3 of those letters in the diagonal. The first of these is already a composite part of the anagram given in the rectangle shown in figure 9, but the second instance is found in the lower right of the bottom panel with respect to the location of the first instance, which may account for the LR fragment in the initial 6 letters. If the diagonal components of these two groupings are extended towards the top panel then they eventually converge in the pt view on the letter I. To put that result in the context of challenge puzzle 1, there is now an implied instance of a 13 stepped pyramid topped by an I. The result already found in 3.1 would strongly suggest that attention is being

directed towards either the Great Seal of the United States or its presence on the US Dollar. With this in mind it is found that extending the center line of the pyramid in the ct view does in fact hit the D of USD in row 1. As a result, this observation is taken as being a key informational output of challenge puzzle 4.

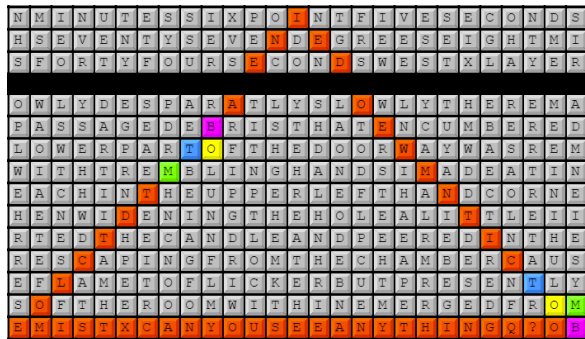


Figure 10: Hidden 13 stepped pyramid formed by the diagonals of the word TOMB in the upper left and lower right.

Rendering this structure in the ct view also reveals that the string ANIT exists along the center line of both the ct and the pt views. The center of the structure in both instances also holds the letters AET and ATE respectively.

3.5 Additional structures

Taking the letters WW from row 10 in the pt and MM from row 4 in the ct it is possible to mark the lower left and upper right hand corners of a 7x7 square that straddles the center line of the pyramid structure perfectly.

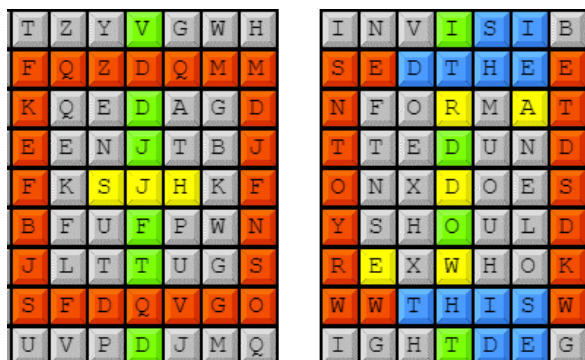


Figure 11: Signatures of Jim Sanborn and Ed Scheidt shown within the 7x7 box formed in columns 13-19 and rows 4-10. The center line of the pyramid structure is shown in green.

The ct view of this structure has at its center the initials SJH, which are taken to be the signature for, “Herbert James Sanborn” and the pt view has a more elaborate scheme as shown in figure 11, whereby the surname Scheidt is anagrammed twice on the outskirts of the 7x7 square whilst the name Edward is given a symmetrical rendition within the inner square.

4 Kinetic elements

It was seen previously in section 3 that a left shift of 13 was required in order to make sense of challenge puzzle 2. Given the ct is laid out in a 32x28 grid with one extra character sitting in a 33rd column on row 23 it is possible to wrap the ct into a cylinder of width 32 by discarding this additional character in row 23, which is deemed an innocuous operation for the purposes of this section as none of the previously identified matrices fall across this row. The same operation can be applied to the tableau panels, which will cause the extra letter L in column 32 to be discarded and for the panels to be wrapped on a width of 31 characters. This difference in width means that the tableau cylinder can actually sit comfortably within the dimensions of the ct and pt cylinders. Giving freedom of movement for all rows about the cylinders axis gives rise to a design that is similar to a Jefferson cylinder. It will be noted that the act of moving individual rows to observe some kind of intended effect is an example of kinetic art, so this section details the outcome of such operations as being distinct from the static information described previously.

4.1 Effects of row shifting

It will be remembered from the previous section that the 3x32 matrix that extended across rows 1-3 was found as a result of challenge 3 to be labelled as matrix #1 and that a left shift of 13 in the first row of this same matrix exposed a third horizontal grouping of F, RE and V. This same shift also causes row one to begin with the letters USD. This initial configuration for matrix #1 will now be used as a starting point to explore the properties of matrix 1 under a defined set of kinetic operations. In this initial configuration, it

is seen that a 4 letter block is present in the ct view composed of 2 vertically occurring digrams, which are given as “UNYY” as shown in figure 12. This string of “UNYY” is deemed to be significant as it starts on the U of the USD alluded to in challenge 4 and most importantly can be seen to encrypt with “OBKR” using the Vigenère tableau to give a value of “KRYP”.

U	S	D	J	K	L	D	K	R	S	H	G	N	F	I	V	J	E	M	U	F	F	H	C	L	R	E	A	X	Y
Y	Q	T	Q	U	X	Q	B	Q	V	U	V	L	L	T	R	E	V	J	Y	Q	T	H	K	S	R	D	H	F	D
V	F	F	S	U	D	E	E	H	S	M	E	T	Z	V	V	S	N	R	K	K	O	E	T	S	F	O	U	N	C

Figure 12: Effect of left shifting 13 characters in the ct view of matrix 1.

It follows immediately that there may be benefit in scanning matrix #1 for the remaining 3 letters, “TOS” that would combine to give a result of “KRYPTOS” for the first 7 letters of K4. Performing such a scan requires finding the letters “CKB” as these will encrypt through the Vigenère tableau to give “TOS”. A left shift of 10 to row 2 allows for such a result and again results in a block of 2 vertically occurring digrams that could be taken at this stage to be either “CKBD” or “CKBF” due to the double occurrence of the letter K in columns 20 and 21.

U	S	D	J	K	Z	L	D	K	R	N	S	H	G	N	F	I	V	J	E	M	U	F	F	H	C	L	R	E	A	X	Y
X	U	V	L	L	T	R	E	V	J	Y	Q	T	H	K	Y	R	D	M	F	C	*	Y	Q	T	Q	U	X	Q	B	Q	V
V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	V	V	G	W	H	K	K	Q	E	T	G	F	Q	J	N	C	E	*

Figure 13: Effect on the ct after left shifting row 2 by 10 characters.

Perhaps the most intriguing outcome of this second operation is, “what has not happened” during the movement of row 2. It can be seen that despite row 2 being left shifted 10 characters, the initial “UNYY” block has remained invariant under the transformation applied. This property of “invariance under transformation” (iut) will be seen to occur consistently in specific transforms of the 6 matrices previously identified including those for which the K4 plain text is already known. Further exploration of the matrix properties after these previous kinetic transformations have been applied shows a number of other phenomena of importance. These observations are best viewed in the accompanying HTML tool, but to recap, all transformations applied to the ct are simultaneously applied to the

pt view. The first 13 character left shift in the pt shows an interesting pattern in which two crosses are formed comprised of the letters HOT and EOF in columns 2 and 22 respectively as shown in figure 14.

S	H	A	D	I	N	S	A	N	D	T	H	E	A	B	S	E	N	C	B	E	F	W	E	E	N	S	U	B	T	L	E
E	S	T	H	E	N	U	A	N	C	E	F	I	O	L	U	S	I	O	N	*	E	O	F	L	I	G	H	T	L	I	E
A	T	W	A	S	T	O	T	A	L	L	Y	I	N	V	I	S	I	B	L	E	R	O	W	S	T	H	A	T	F	O	*

Figure 14: Corresponding effect in the pt view of matrix when left shifting 13 characters.

This pattern in the pt is destroyed after the second transformation in which row 2 is left shifted 10 characters, but interestingly a complete 4 letter block for the letters “THEO” is now found as a transformed quantity.

S	H	A	D	I	N	S	A	N	D	T	H	E	A	B	S	E	N	C	B	E	F	W	E	E	N	S	U	B	T	L	E
E	S	T	H	E	N	U	A	N	C	E	F	I	O	L	U	S	I	O	N	*	E	O	F	L	I	G	H	T	L	I	E
A	T	W	A	S	T	O	T	A	L	L	Y	I	N	V	I	S	I	B	L	E	R	O	W	S	T	H	A	T	F	O	*

Figure 14: Transformation of the pt matrix when a left shift of 10 is applied to row 2.

Analyzing which letters remain in proximity before and after the first and second transformation it is further seen that the letters F and I are also implicated when considering the before and after case, meaning that an entire 6 character block has remained invariant. The relevance of this observation will be discussed below, but it will now be shown that this invariance principle is given a full exposition by way of an actual embedded example in the pt.

4.3 Invariance under transformation

It can readily be calculated that the number of distinct row combinations for a 3x32 matrix can be computed as 32x32x32, giving a total figure of 32,768 possible row permutations. Scanning for the numbers 3, 2, 7, 6 and 8 in the pt it is found that all five numbers occur in proximity within the bounds of a 3x32 matrix over rows 12-14. Furthermore, they can be re-arranged through row shifting to give a visual exposition of the invariance phenomena already shown in 4.2. Note that these operations are applied to the corrected pt matrix containing XLAYERTWO on row 14 rather than IDBYROWS. Left shifting rows 12, 13 and 14 by 3 characters will help keep the

number EIGHT in view during the second transformation step. If row 14 is now right shifted 13 characters it will be seen in figure 15 that the number TWO now sits underneath the numbers SIX and SEVEN. The number THREE will be seen in figure 15 to be present in a 5 letter cross in column 31. If row 2 is now right shifted by 7 characters then the numbers themselves can be seen to have remained unaltered, but there have been subtle changes to their positions. Another artefact of the transformation is the emergence of the letters F, RE and V shown in yellow in figure 15.

M	I	N	U	T	E	S	S	I	X	P	O	I	N	T	F	I	V	E	S	E	C	O	N	D	S	N	O	V	E	N			
S	E	V	E	N	T	S	E	V	E	N	D	E	G	R	E	E	S	E	I	G	H	T	M	I	N	U	T	E	S	S	I	X	
T	X	L	A	Y	E	R	F	M	O	T	E	S	F	O	R	T	F	O	U	R	S	E	C	O	N	D	S	E	S	E	S	I	X

M	I	N	U	T	E	S	S	I	X	P	O	I	N	T	F	I	V	E	S	E	C	O	N	D	S	N	O	V	E	N																								
I	N	U	T	E	S	S	I	X	P	O	I	N	T	F	I	V	E	S	E	C	O	N	D	S	E	S	E	S	I	X	P	O	I	N	T	F	I	V	E	S	E	C	O	N	D	S	N	O	V	E	N			
T	X	L	A	Y	E	R	F	M	O	T	E	S	F	O	R	T	F	O	U	R	S	E	C	O	N	D	S	E	S	E	S	I	X	P	O	I	N	T	F	I	V	E	S	E	C	O	N	D	S	N	O	V	E	N

Figure 15: Exposition of “invariance under transformation” for the numbers 32768 in the 3x32 matrix bound by rows 12-14 of the pt. Effect of a 7 character right shift of row 2 is shown directly below.

This kinetic exposition of the iut principle that has already seen in matrix #1 is deemed to be important as it is reinforcing the idea that a 3x32 matrix may have two positions that can be compared in which the transformation is effected solely by a movement of the middle row.

4.4 Symmetry operations

Before arriving at some concrete conclusions as to the mechanics of the K4 cipher, some additional observations can be made about the previous behaviour of matrix #1 under transformation. Each row and by extension any 3x32 matrix in the ct and pt has a symmetrical opposite, whose row can be defined as $29 - n$, where n is the row to be operated on. This means that matrices 1-6 all have symmetrical counterparts, with the exception of matrix 5, which straddles rows 14 and 15 unevenly and forms an exception case. Every operation is also mirrored in the pt, so simultaneous operations across 4 matrices are made every time a single row movement is made under such a paradigm. This idea of operating on 4 matrices simultaneously echoes the setup of the initial challenge puzzles in which 4 distinct 7x6

matrices had to be considered in order to make progress. If the row shifts on matrix 1 made previously in 4.2 are mirrored in the pt using a counter-symmetric rotation scheme then rows 1-3 and 26-28 end up with the configuration shown in figure 17. The letters ANL are present in both the original and mirrored matrix and are shown in red with their diagonals extended to show that they cross above the letters W and L in row 15.

S	H	A	D	I	N	G	S	N	D	T	H	E	A	B	E	S	N	C	B	E	T	W	E	N	S	S	E	T	L	E		
E	S	T	H	E	N	D	A	C	E	O	F	I	C	O	N	S	I	O	N	E	O	F	F	L	I	G	H	T	E	D		
E	T	W	A	S	T	O	R	A	F	L	Y	I	N	V	E	S	I	B	L	E	H	O	W	S	T	H	A	T	P	O		
E	S	I	B	L	E	T	H	E	V	U	S	E	D	T	H	E	E	A	R	T	H	S	M	A	G	N	E	T	*	*		
A	C	F	I	E	L	D	X	T	H	E	I	N	F	O	R	M	A	T	I	O	N	W	A	S	G	A	T	H	E	R		
S	E	D	A	N	D	T	R	A	N	S	M	I	T	T	E	D	U	N	D	E	R	G	R	U	N	D	T	O	A	N		
N	K	N	O	W	N	O	C	A	T	I	O	N	X	D	O	E	S	L	A	N	G	L	E	Y	K	N	O	W	A	*		
R	O	U	T	H	I	S	E	T	H	E	Y	S	E	T	H	E	I	S	E	T	H	E	R	O	U	T	H	E	O	*		
T	H	E	R	E	S	O	N	E	W	H	E	R	E	M	W	H	O	K	N	O	W	S	T	H	E	E	X	A	C	T		
O	C	A	T	I	O	N	T	O	N	L	Y	N	W	N	T	H	I	S	W	A	S	H	I	S	L	A	S	T	M	E	S	*
S	A	G	E	X	T	H	I	R	T	Y	E	I	G	H	T	D	E	G	R	E	E	S	F	I	F	T	Y	S	E	*		
V	E	N	M	I	N	U	T	E	S	S	I	X	P	O	I	N	T	F	I	V	E	S	E	C	O	N	D	S	N	O	*	
R	T	H	S	E	V	E	N	T	S	E	V	E	N	D	E	G	R	E	E	S	E	I	G	H	T	M	I	N	U	*		
R	E	S	E	C	O	N	D	E	F	O	U	R	S	E	C	O	N	D	S	E	S	E	S	I	X	L	A	Y	E	R	W	O

S	L	O	W	L	Y	E	S	P	A	R	A	T	I	V	S	L	O	W	L	Y	T	H	E	R	E	M	A	I	N	S				
O	F	P	A	S	S	A	G	E	D	E	B	R	I	S	T	H	A	T	E	N	C	O	N	U	M	B	E	R	E	D	T	*		
A	E	L	O	W	E	R	E	A	R	T	O	F	T	H	E	D	O	O	R	W	A	Y	W	A	S	R	E	M	O	V	*			
E	D	W	I	T	H	E	R	E	M	B	L	I	N	G	S	H	A	N	D	E	A	T	I	N	Y	*	*	*	*	*				
R	E	A	C	H	I	N	T	R	E	U	P	E	R	E	L	E	T	H	A	N	D	O	C	O	N	S	E	R	A	N	*			
D	T	H	E	N	W	I	D	E	N	I	N	G	T	H	E	H	O	L	E	A	L	T	I	L	E	S	I	N	*	*				
S	E	R	T	E	D	T	H	E	C	A	N	D	L	E	A	N	D	D	E	E	R	E	D	I	N	T	H	E	O	*				
A	I	R	E	S	C	A	P	I	N	G	F	R	O	M	T	H	E	C	H	A	M	B	E	R	C	A	U	S	E	D	*			
T	H	E	F	L	A	M	E	T	O	F	L	I	C	K	E	R	B	U	T	P	R	E	S	E	N	T	L	Y	D	E	*			
T	L	S	O	F	T	H	E	R	O	O	N	W	I	T	H	I	N	E	M	E	R	G	E	D	F	R	O	M	*	*				
T	H	E	M	I	S	T	X	C	A	N	Y	O	U	R	E	S	C	A	N	Y	T	H	E	R	I	N	G	Q	T	O	B	K	R	*
V	O	X	O	G	H	T	E	S	O	L	I	F	E	B	E	N	F	L	R	V	Q	Q	P	R	N	G	R	S	S	O	*			
A	W	I	N	F	B	O	N	E	P	T	W	I	Q	S	V	Q	S	E	K	E	Z	Z	W	A	T	J	L	U	D	I	*			
O	K	U	H	U	C	H	E	K	C	A	R	*	V	T	Z	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	*		

Figure 17: Inferred ray projected above second instance of WL corresponding to the letters YA and R in the ct with the position of R shown in yellow as D in the pt view.

The act of joining these two paths for the ANL trigram in the original and mirrored matrices exposes the presence of a second kinetic scheme that is outside the scope of this document to explain fully, but the outcome of the operations it facilitates once followed through to conclusion explain the presence of the raised letters YA and R in the ct. In short, the blue lines in figure 17 are taken to be metaphoric of light rays passing through focal points defined at the positions of the WL characters, which have a separation width of 16. This means that the 2 instances of WL can continually be swapped by rotating the cylinder row 180° each time.

4.5 The Kryptos alphabet

A combination of brute force computation and the solution to the second kinetic scheme very briefly

described above eventually found that each matrix defined in the ct supports an additional configuration, which is found by rotating the mirrored matrices in a symmetric rather than counter symmetric orientation. This operation alone is not enough to mix the digraph statistics that are being drawn upon by the cipher scheme significantly enough to offer much difference to the original matrix configuration, but it is reasoned that every time 8 letters have been selected from a given matrix the entire ct and pt are all moved one position right through the Kryptos alphabet. This combination of difference in symmetry operation combined with the statistical mixing of the diagram relationships is enough to give markedly different results for the optimal solutions to the first and second configuration positions. The overall effect of such a scheme in which each matrix position supports two configurations is that 12 distinct matrices can be defined that each result in the output of 8 letters. The first configuration for matrix #1 has no Kryptos right shift applied to the ct or pt, but by the time the 12th matrix configuration is reached it will be seen that right shifting has been applied 11 times. These 12 matrices each outputting 8 letters will of course deliver 96 letters of plain text leaving only the final “R” of K4 unmapped.

5 Cymatic behaviour

In the course of this paper the proposed mechanics of the K4 cipher have slowly been built from first principles by following guidance from a series of static and kinetic challenges that have each revealed additional information at each step. Using the HTML tool that accompanies this paper it can be seen that the scheme outlined generates all known pt for K4 in the correct positions, but another full paper could be written on what is happening in cryptograms 4 and 7 where the full plaintext is already known. Contrasting the matrix configurations for cryptograms 4 and 7 it is readily apparent that there are two more phenomena of interest. The first is that the number of rows being considered for the initial

position of any matrix configuration will always contain a permutation of ABC in the vertical plane or diagonal plane, which drastically cuts down the initial row permutations that need to be considered. This requirement would appear to change after cryptogram 6 at which point the requirement becomes XYZ. The most significant observation, which can also be seen in the operations already shown for matrix #1 is that all 4 views participating in any given solution show a curious reinforcement of positional information relating to digrams that occur in whole blocks of 4 characters. The 8 letters for any given solution are always extracted from the original matrix position in the ct, but the 3 corresponding matrices all appear to show a range of different relationships to the positions from which these characters are extracted from the primary matrix. This phenomenon is akin to a form of interference occurring between the diagram statistics of each matrix view that changes each time the middle row is moved by one character in a transformation operation. For that reason it is conjectured that the underlying mathematical mechanism of the K4 cipher is actually a maximization function of some description that defines a starting and finishing configuration in which both invariance under transformation and positional statistics both give weight to whether a given row transformation is indeed the intended finishing configuration.

5.1 Further research

A candidate solve that satisfies the base constraints outlined for the K4 scheme described in this paper has already been extracted for 96 letters that combine with the unmapped R in position 97 to give the following result:

KRYPTO SITE AM EAST. ROUND EAST
NORTH EAST. I FORCES. DYNAMO WAVE
ROUND GROUP. TE. BERLIN CLOCK.
CHURCH TUNNEL. UK GCHQ CODE[R].

This result was achieved via visual inspection, but new effort is currently underway to explore the roles of automated functions that can keep score of maximum and minimum invariance.